Healthcare Monitoring Device Network

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DSA PROJECT

Overview:

This project implements a Healthcare Monitoring Device Network using the Graph Data Structure in C. It simulates how medical devices (like heart monitors, oxygen sensors, etc.) are connected and managed within a hospital network.
The system supports full CRUD operations (Create, Read, Update, Delete) on devices and allows visualization of their network connections using an adjacency matrix.

Features:

- Add new medical devices (Create)
- View all registered devices (Read)
- Update device details (Update)
- Delete devices from the network (Delete)
- Create connections between devices
- Display network graph (Adjacency Matrix)

Data Structures Used:

- Structure (struct) → To store device details
- Graph (Adjacency Matrix) → To represent connections
- Arrays → For storing devices and network relationships

Technologies:

- Language: C
- Compiler: GCC / Turbo C / Any standard C compiler
- Concept: Graph Theory + CRUD Operations

How It Works:

Each device contains:

- ID → Unique identifier
- Name → Device type (e.g., ECG, BP Monitor)
- Status → Active / Inactive / Fault
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Connections between devices are stored in an adjacency matrix, where:

- 1 = Connected
- 0 = Not connected

How to Run:

1.Copy the code into a file:

healthcare_network.c

2.Compile the program:

```
gcc healthcare_network.c -o network
```

3. Run the executable:

```
./network
```

Sample Menu:

--- Healthcare Device Network ---

1. Add Device
2. View Devices
3. Update Device
4. Delete Device
5. Add Connection
6. View Network
7. Exit

Example Use Case:

- Add devices like:
- Heart Monitor
- Oxygen Sensor
- Temperature Sensor
- Connect them to simulate a hospital ICU network
- Monitor device status and connections

Future Enhancements:

- Implement BFS/DFS traversal
- Add file storage (persistent data)
- Real-time alert system
- Convert into GUI application
- Database integration

THANK YOU